UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Implicit and Composite Functions

Complete study pack for class 004: level surfaces and the differential, implicit differentiation in two and three variables, composite functions and the chain rule, and polar-coordinate problems

Compiled from the PESU class 004 slide deck and its two AV-summary transcripts, the implicit-function problems that close class 003, the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topics 1–3 follow the first class 004 lecture (implicit functions); topics 4–5 follow the problem session on composite functions. The same material also appears, together with class 003, in the pack "Total Derivative, Chain Rule and Implicit Functions"; this pack is the focused class 004 version. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9 · S003·26	Deck 004 / 003, slide number (rendered .pptx)
T004a 00:13:09	Transcript 004 part I: "Implicit functions and problems"
T004b 00:02:31	Transcript 004 part II: "Problems on composite and implicit functions"
T003b 00:10:33	Transcript 003 part II (implicit problems at its end)
CW U1 P17 · ESA Jan-24 s3 1d	Unit 1 class-work problem 17 · past-paper question

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6. Appendix A · Symbol glossary
7. Appendix B · Sources, coverage and gaps

1 Implicit functions, level surfaces and the differential

A relation $f(x,y)=c$ (or $f(x_1,\dots,x_n)=c$) defines a function implicitly; geometrically it is a level curve or level surface, on which f keeps the same value. Examples are equipotential surfaces and isotherms. On a level surface the total differential $df = f_x\,dx + f_y\,dy + \dots$ is zero, and that single fact gives every implicit-differentiation formula. The differential is also used in substitution and error estimates.

Key terms and definitions

Implicit function. A function given by a relation $f(x,y)=c$ (for example $x^3y+3xy^2=3$) rather than by a formula $y=g(x)$. S004·3 · T004a 00:00:18

Level surface (level curve). The set where $z=f(x,y)=c$ (or $f(x_1,\dots,x_n)=c$). Examples: equipotential surfaces, isotherms. The cylinders $x^2+y^2=c$ and the spheres $x^2+y^2+z^2=c$ are layers, one for each c. S004·4 · T004a 00:02:35 · T004a 00:03:29

Total differential. For $z=f(x,y)$: $dz=f_x\,dx+f_y\,dy$. For $z=f(x_1,\dots,x_n)$: $dz=f_{x_1}dx_1+\dots+f_{x_n}dx_n$. The total derivative is also called the differential. S004·3 · S004·4 · T004a 00:02:04

> *"The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation."* S004·4

> *"If z represents a level surface, i.e. if $z=f(x,y)=c$, then $dz=0$. The total derivative is also called the differential."* S004·4

Formulas

E1.1 · Total differential ESA ★★

$$dz = f_x(x,y)\,dx + f_y(x,y)\,dy$$

gives	Approximate change in z for small changes dx, dy.
applies	$z=f(x,y)$ differentiable.
note	n variables: $dz=f_{x_1}dx_1+f_{x_2}dx_2+\dots+f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

Graphs and diagrams

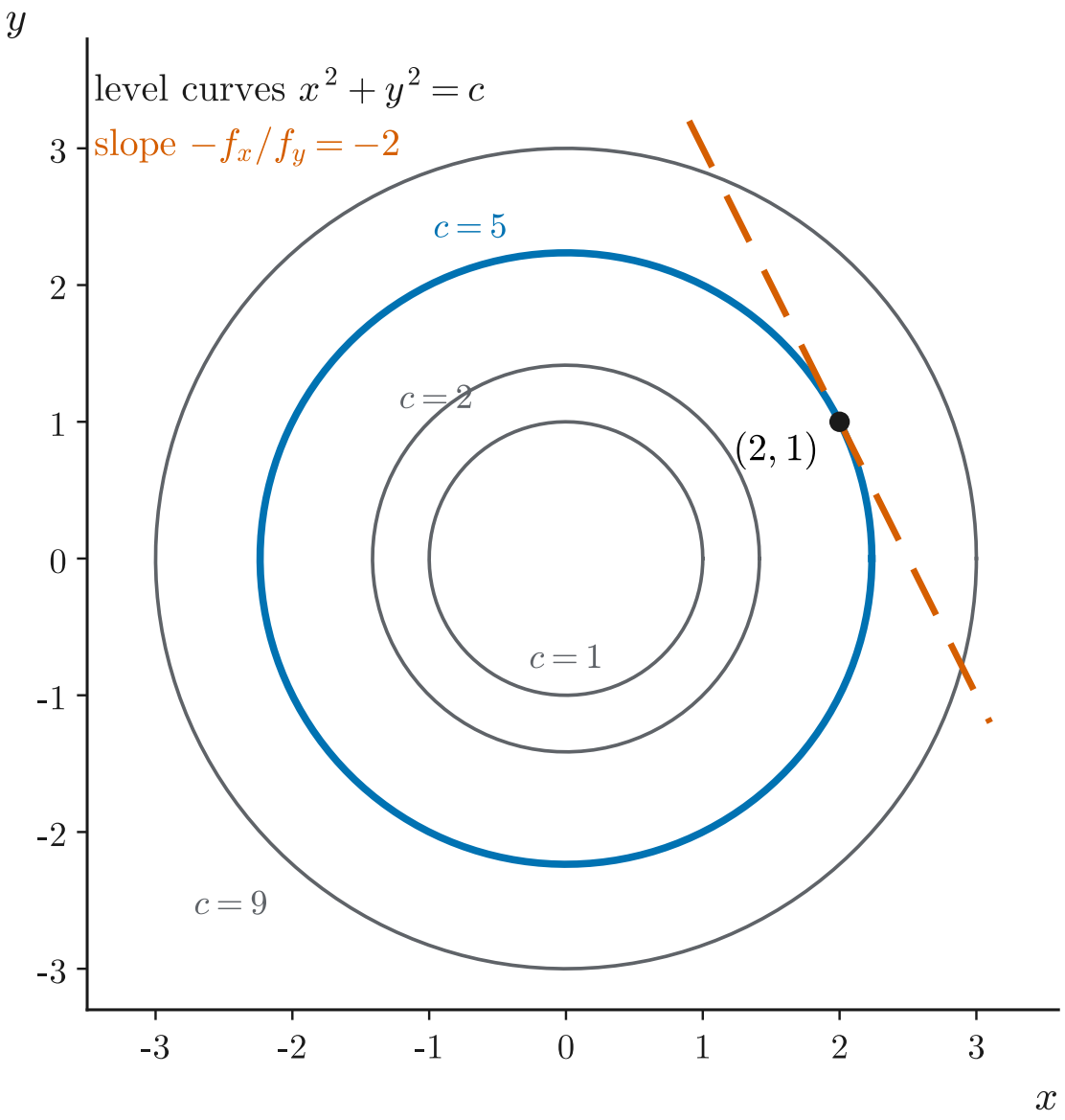


Figure G1.2. Level curves and the slope $-f_x/f_y$. ESA ★★

ADDED FOR CLARITY S004·4 · S003·7 · T004a 00:03:10

How to read it. Level curves of $f = x^2 + y^2$ for $c = 1, 2, 5, 9$: the cross-sections of the cylinders on slide 004·4. On $c = 5$ at $(2, 1)$ the dashed tangent has slope $-f_x/f_y = -2x/2y = -2$, the same $dy/dx = -x/y$ you get by differentiating $x^2 + y^2 = 5$ directly.

$x^2 + y^2 + z^2 = c$: one shell per value of c

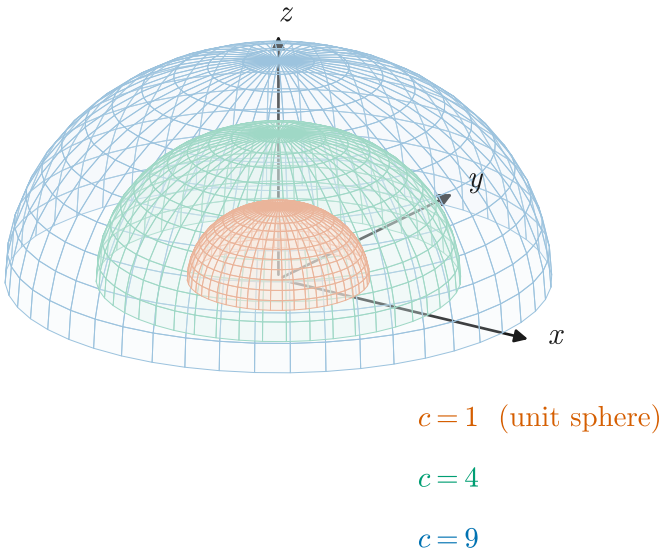


Figure G1.1. Level surfaces: spheres $x^2 + y^2 + z^2 = c$. ESA ★

S004·4 · T004a 00:03:29 · T004a 00:03:42

How to read it. Each shell is one level surface: on it $x^2 + y^2 + z^2$ has the same value c (only the upper halves are drawn, as on the slide). Larger c gives a larger sphere and $c = 0$ gives the origin. Think of each shell as one isotherm or equipotential surface. Moving along a shell does not change f, so $df = 0$ there.

Important points and why

1. Integration by substitution uses the differential: with $t = x^2$, $dt = 2x\,dx$, so $\int e^{x^2} x\,dx = \frac{1}{2}\int e^t\,dt$. ESA ★ TRANSCRIPT ONLY T004a 00:07:57 · T004a 00:08:29
2. Errors and approximations, another application of the differential, come at the end of the unit. ESA ★ TRANSCRIPT ONLY T004a 00:08:47
3. Restriction to remember: the formula needs $z = f(x, y)$ to be a constant (a level curve). Move everything to one side first: $f(x, y) = c$ or $f(x, y) = 0$. ESA ★★★ T004a 00:01:49 · T004a 00:02:04
4. Level surfaces in physics: equipotential surfaces (constant potential) and isotherms (constant temperature). ESA ★ T004a 00:02:35
5. $x^2 + y^2 = c$ gives a family of cylinders (layers), and $x^2 + y^2 + z^2 = c_1$ a family of spheres about the origin: $c_1 = 0$ is the origin, $c_1 = 1$ the unit sphere, larger c_1 larger spheres. ESA ★ S004·4 · T004a 00:03:10 · T004a 00:03:42

6. $f(x_1, \ldots, x_n) = c$ is still a level surface for $n > 3$, even though it cannot be plotted; its total differential is still 0. ESA ★ T004a 00:04:34

SHORT-NOTE RECAP

A relation $f(x, y) = c$ or $f(x_1, \ldots, x_n) = c$ defines an implicit function; geometrically it is a level curve or level surface (isotherm, equipotential surface), e.g. the cylinders $x^2 + y^2 = c$ or spheres $x^2 + y^2 + z^2 = c$. On it f does not change, so its total differential $df = f_x\,dx + f_y\,dy + \cdots = 0$. The differential is also used in substitution in integrals and in error estimates.

2 Implicit differentiation in two variables

For $f(x,y)=c$, y is a function of x and $\dfrac{dy}{dx}=-\dfrac{f_x}{f_y}$. The branch diagram shows why. The formula replaces long product-rule differentiation, gives tangent lines through the slope, and handles relations such as $(\cos x)^y=(\sin y)^x$. Combined with a composite function, it gives du/dx along a curve.

Key terms and definitions

> *"Setting* $\dfrac{dz}{dx}=0$ *leads to a simple computation formula for implicit differentiation:* $f_x\cdot 1+f_y\dfrac{dy}{dx}=0$." S004·5

Formulas

E2.1 · Implicit differentiation, two variables ESA ★★

$$f(x,y)=c \implies \frac{dy}{dx}=-\frac{f_x}{f_y}, \qquad f_y\neq 0$$

GIVES	Slope of the level curve at a point.
APPLIES	$f_y\neq 0$ at the point.
NOTE	From $df=f_x\,dx+f_y\,dy=0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E2.2 · Tangent line at a point ESA ★★

$$y-y_0=m\,(x-x_0), \qquad m=\left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

GIVES	Equation of the tangent to the curve at (x_0,y_0).
APPLIES	(x_0,y_0) on the curve.
NOTE	Through the origin: $y=mx$.

S004·7 · T004a 00:14:25

E2.3 · Derivative of a constant power and a variable power ESA ★★

$$\frac{d}{dx}a^x=a^x\ln a, \qquad \frac{d}{dx}x^n=n\,x^{n-1}$$

GIVES	The two rules needed for $(\cos x)^y$ and $(\sin y)^x$.
APPLIES	In a partial derivative, the other variable is a constant.
NOTE	$\ln$ needs a positive argument: $\cos x>0$, $\sin y>0$.

S004·8 · T004a 00:16:43

Graphs and diagrams

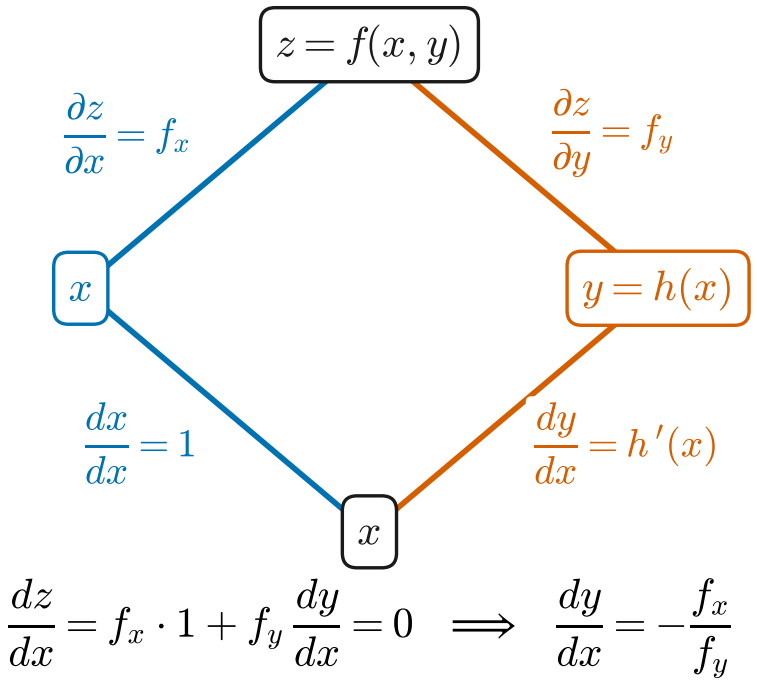


Figure G2.1. Branch diagram for implicit differentiation. ESA ★★

S004·5 · T004a 00:07:57

How to read it. Differentiate $z = f(x, y)$ with respect to x along both branches. The left branch is direct ($f_x \cdot dx/dx = f_x \cdot 1$). The right branch goes through $y = h(x)$ ($f_y \cdot dy/dx$). On a level curve the total is 0: $f_x + f_y \, dy/dx = 0$.

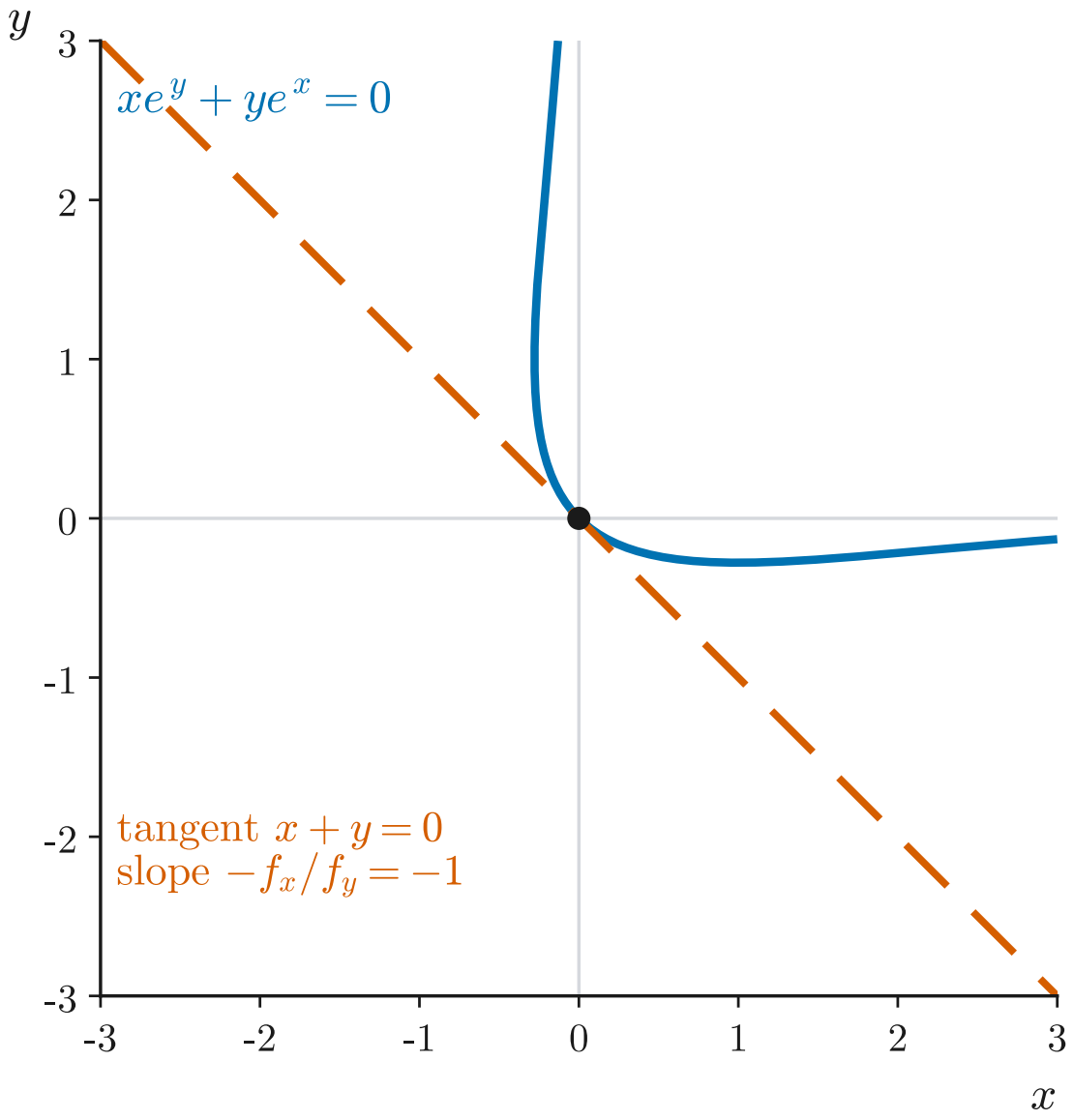
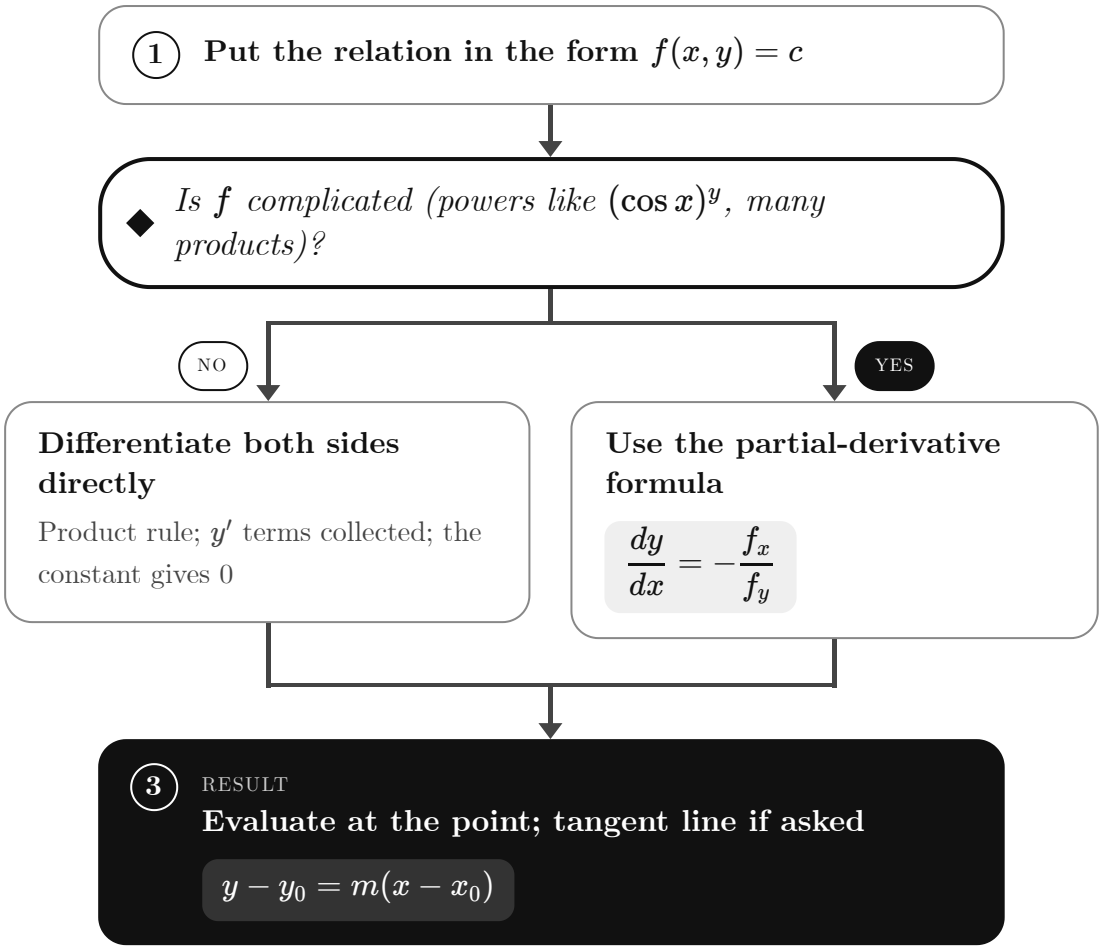


Figure G2.2. The curve $xe^y + ye^x = 0$ and its tangent at the origin. ESA ★★

S004·7 · T004a 00:13:48

How to read it. The blue curve is the set where $f(x, y) = xe^y + ye^x = 0$. At the origin $dy/dx = -f_x/f_y = -1$, so the dashed tangent is $y = -x$. The curve is plotted from its equation (the slide's picture is a 3-D grapher screenshot).

Flowcharts



Remark. Both routes give the same dy/dx (W2.1, W2.2). The formula needs $f_y \neq 0$ at the point.

Figure F2.1. Finding dy/dx for an implicit function: two routes. ESA ★★

S003·27 · S004·3 · S004·5 · T003b 00:10:33 · T004a 00:06:19

Important points and why

1. Differentiating an implicit relation directly needs the product rule on every term, and the right-hand constant gives 0. ESA ★★ S004·3 · T004a 00:00:48
2. The formula $dy/dx = -f_x/f_y$ gives the same answer and is far easier when f is complicated. ESA ★★ T004a 00:06:19 · T004a 00:06:36
3. There are two ways to solve "find dy/dx": differentiate both sides directly, or use $-f_x/f_y$. Both are accepted. ESA ★★ S003·27 · T003b 00:10:33
4. The slope dy/dx at a point gives the tangent line $y - y_0 = m(x - x_0)$. ESA ★★ S004·7 · T004a 00:12:01
5. For $(\cos x)^y = (\sin y)^x$ the result needs $\cos x > 0$ and $\sin y > 0$, because of the logarithms. ESA ★ T004a 00:18:12
6. Implicit and composite together: if $u = \tan(x^2 + y^2)$ and $x^2 - y^2 = 2$, use the relation to eliminate y (or find dy/dx) before differentiating. ESA ★★ CW U1 P16 · CW U1 P18

Why is $dy/dx = -f_x/f_y$?
Along the curve $f(x,y) = c$ the value of f does not change, so $df = f_x\,dx + f_y\,dy = 0$. Divide by dx: $f_x + f_y\,\dfrac{dy}{dx} = 0$. The branch diagram (Figure G2.1) shows the same sum.
S004·4 · S004·5 · T004a 00:07:39

Why is the level-curve slope $-f_x/f_y$ and not $-f_y/f_x$ (as printed on slide 004·4)?
From $f_x + f_y\,y' = 0$ you get $y' = -f_x/f_y$. Test it on $x^2 + y^2 = 5$ at $(2,1)$: $-f_x/f_y = -2x/(2y) = -2$, which matches differentiating directly; $-f_y/f_x$ would give $-1/2$.
ADDED FOR CLARITY S004·4 · S004·5

Exam trap. Slide 004·4 prints $dy/dx = -f_y(x,y)/f_x(x,y)$. The correct formula is $dy/dx = -f_x/f_y$. ADDED FOR CLARITY S004·4

Exam trap. In $(\cos x)^y$, with respect to x it is a power ($y(\cos x)^{y-1}(-\sin x)$); with respect to y it is an exponential ($(\cos x)^y \ln\cos x$). S004·8 · T004a 00:16:02

Worked examples

W2.1 · Direct differentiation vs the formula — ESA ★★

GIVEN $x^3y + 3xy^2 = 3$.

STEP 1 Direct: $x^3y' + 3x^2y + 3y^2 + 6xy\,y' = 0$.

STEP 2 Formula: $f_x = 3x^2y + 3y^2$, $f_y = x^3 + 6xy$.

STEP 3 $y' = -\dfrac{3x^2y + 3y^2}{x^3 + 6xy}$.

ANSWER $\boxed{\dfrac{dy}{dx} = -\dfrac{3y(x^2+y)}{x(x^2+6y)}}$

CHECK Both methods agree; the slide's printed numerator is unclear.

S004·3 · T004a 00:00:48 · T004a 00:05:41

W2.2 · Two ways for $x^2 + xy + y^2 = 7$ — ESA ★★

GIVEN $x^2 + xy + y^2 = 7$.

STEP 1 Direct: $2x + (xy' + y) + 2y\,y' = 0$, so $(x+2y)y' = -(2x+y)$.

STEP 2 Formula: $f_x = 2x + y$, $f_y = x + 2y$, $y' = -f_x/f_y$.

ANSWER $\boxed{\dfrac{dy}{dx} = -\dfrac{2x+y}{x+2y}}$

CHECK Same by both methods.

S003·27 · T003b 00:10:23

W2.3 · Tangent line at the origin ESA ★★

GIVEN $xe^y + ye^x = 0$; tangent at $(0,0)$.

STEP 1 $f_x = e^y + ye^x$, $f_y = xe^y + e^x$.

STEP 2 $\dfrac{dy}{dx} = -\dfrac{e^y + ye^x}{xe^y + e^x}$; at $(0,0)$: $-\dfrac{1+0}{0+1} = -1$.

STEP 3 $y - 0 = -1(x - 0)$.

ANSWER $\boxed{x + y = 0}$

CHECK $(0,0)$ satisfies the equation.

S004·7 · T004a 00:11:48

W2.4 · $(\cos x)^y = (\sin y)^x$ ESA ★★

GIVEN $f(x,y) = (\cos x)^y - (\sin y)^x = 0$.

STEP 1 $f_x = y(\cos x)^{y-1}(-\sin x) - (\sin y)^x \ln \sin y$.

STEP 2 $f_y = (\cos x)^y \ln \cos x - x(\sin y)^{x-1}\cos y$.

STEP 3 $y' = -f_x / f_y$. Divide through by $(\cos x)^y = (\sin y)^x$, then multiply top and bottom by $\cos x \sin y$.

ANSWER $\boxed{\dfrac{dy}{dx} = \dfrac{\sin y\,(y \sin x + \cos x \ln \sin y)}{\cos x\,(\sin y \ln \cos x - x \cos y)}}$

CHECK Checked numerically at a point on the curve ($x = 0.5$, $y \approx 0.909$): both give -0.4980.

S004·8 · T004a 00:14:57

W2.5 · Slope at (1, 1) ESA ★★

GIVEN $e^y - e^x + xy = 1$ at $(1,1)$.

STEP 1 Check: $e - e + 1 = 1$, so the point is on the curve.

STEP 2 $f_x = -e^x + y$, $f_y = e^y + x$.

STEP 3 $y' = -\dfrac{-e+1}{e+1}$.

ANSWER $\boxed{\dfrac{dy}{dx} = \dfrac{e-1}{e+1}}$

CHECK CW answer: $(e-1)/(e+1)$.

CW U1 P17

W2.6 · Composite with an implicit constraint ESA ★★

GIVEN $u = \tan(x^2 + y^2)$ with $x^2 - y^2 = 2$; find du/dx.

STEP 1 From the relation, $y^2 = x^2 - 2$, so $u = \tan(2x^2 - 2)$.

STEP 2 $\dfrac{du}{dx} = \sec^2(2x^2 - 2) \cdot 4x.$

ANSWER $\boxed{\dfrac{du}{dx} = 4x \sec^2(2x^2 - 2)}$

CHECK Via the chain rule: $u_x + u_y\, y'$ with $y' = x/y$ gives the same. CW answer agrees.

CW U1 P16

W2.7 · Total derivative along an implicit curve ESA ★★

GIVEN $u = x \log(xy)$ with $x^3 + y^3 - 3xy - 1 = 0$; find du/dx.

STEP 1 $\dfrac{du}{dx} = u_x + u_y \dfrac{dy}{dx}$, with $u_x = \log(xy) + 1$, $u_y = \dfrac{x}{y}$.

STEP 2 From the relation: $\dfrac{dy}{dx} = -\dfrac{3x^2 - 3y}{3y^2 - 3x} = \dfrac{y - x^2}{y^2 - x}$.

ANSWER $\boxed{\dfrac{du}{dx} = 1 + \log(xy) + \dfrac{x\,(y - x^2)}{y\,(y^2 - x)}}$

CHECK CW answer agrees; sympy residual 0.

CW U1 P18

SHORT-NOTE RECAP

An implicit function is given by a relation $f(x, y) = c$, a level curve. Along it $df = f_x\, dx + f_y\, dy = 0$, so $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$ $(f_y \neq 0)$. Direct differentiation of both sides gives the same result. The slope at (x_0, y_0) gives the tangent $y - y_0 = m(x - x_0)$; e.g. $xe^y + ye^x = 0$ has tangent $x + y = 0$ at the origin.

3 Implicit functions of three variables

A relation $f(x,y,z)=0$ is a level surface and makes z a function of x and y. Holding y fixed gives $\partial z/\partial x = -f_x/f_z$. Holding z fixed gives $\partial y/\partial x = -f_x/f_y$. Denominators must be non-zero, the point must lie on the surface, and second derivatives must remember that z still depends on x and y.

Key terms and definitions

Implicit function of three variables. $f(x,y,z)=0$ with z (or y) regarded as a function of the remaining variables; the other variables are intermediate variables. S004·6 · T004a 00:09:05

> *"Let $f(x,y,z)=0$ be the equation of an implicit function of three variables x,y,z. Suppose y and z are functions of x, then f is a function of one independent variable x and y, z are intermediate variables."* S004·6

Formulas

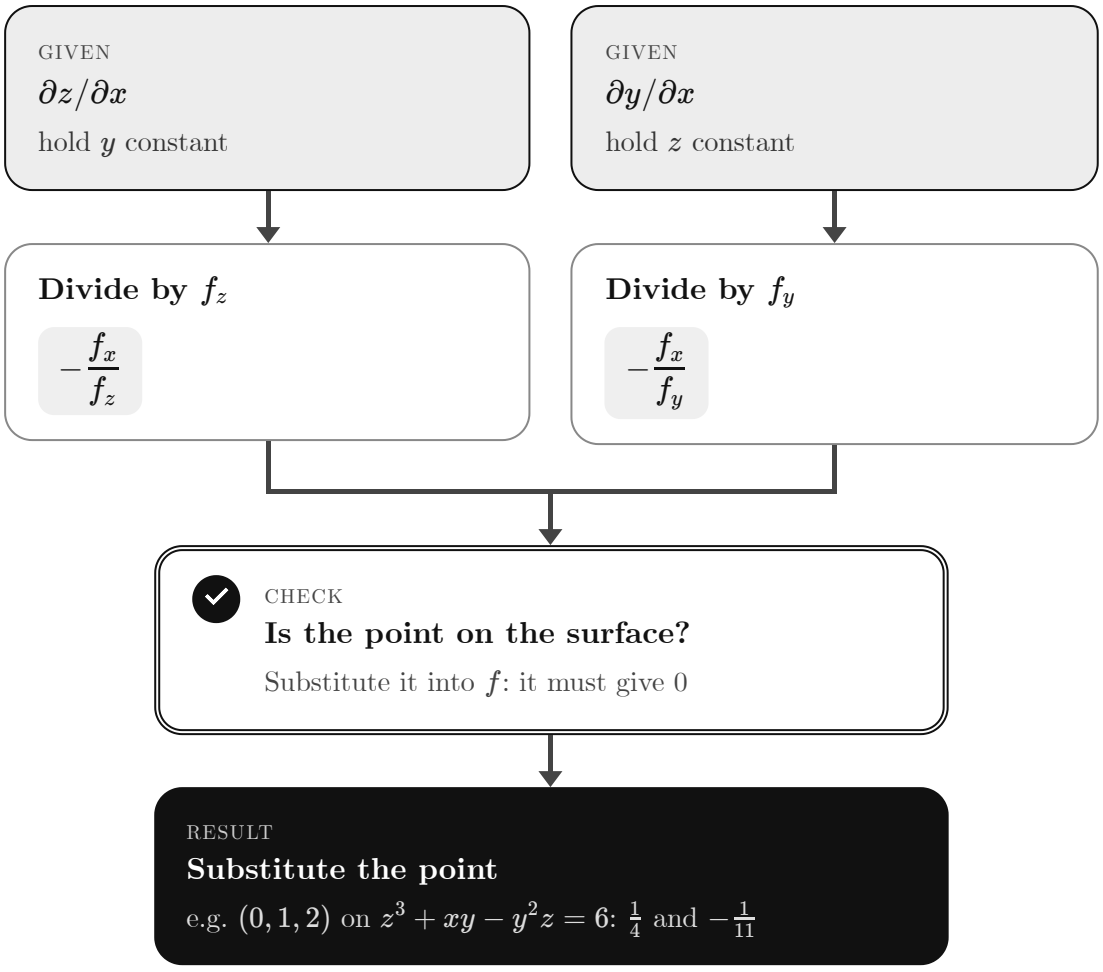
E3.1 · Implicit differentiation, three variables ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

gives	Partial derivatives of the implicitly defined variable.
applies	Hold y constant for z_x; hold z constant for y_x.
note	Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

Flowcharts



Exam trap. The lecture used $(1,0,2)$, which is not on the surface; always check the point first.

Figure F3.1. Partial derivatives from $f(x,y,z)=0$. ESA ★★

S004·6 · S004·9 · S003·26

Important points and why

1. For $f(x,y,z)=0$, x and y are independent, so dz alone has no meaning; find the partials z_x, z_y. ESA ★★ T004a 00:09:38
2. For $\partial z/\partial x$ treat y as constant; for $\partial y/\partial x$ treat z as constant. ESA ★★ S004·6 · S004·9 · T004a 00:19:00
3. The formula avoids separating z from a messy equation such as $x^3y^2+3xyz+\sin(e^z)=c$. ESA ★ T004a 00:10:55
4. Check that the given point satisfies $f=0$ before substituting. ESA ★★ (added for clarity) S004·9 · T004a 00:18:39

Exam trap. The lecture works problem 3 at $(1,0,2)$ and gets 0 for both derivatives, but $(1,0,2)$ is not on $z^3+xy-y^2z=6$ (it gives 8). Use the slide's point $(0,1,2)$: $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$. ADDED FOR CLARITY S004·9 · T004a 00:20:47

Exam trap. Do not differentiate $-f_x/f_z$ as if z were a constant when you need second derivatives: z is still a function of x and y (W3.3). ADDED FOR CLARITY ESA Jan-24 s3 1d

Worked examples

W3.1 · $\partial y/\partial x$ and $\partial z/\partial x$ at $(0,1,2)$ — ESA ★★

GIVEN $z^3+xy-y^2z=6$ at $(0,1,2)$.

STEP 1 Check the point: $8+0-2=6$.

STEP 2 $f=z^3+xy-y^2z-6$: $f_x=y$, $f_y=x-2yz$, $f_z=3z^2-y^2$.

STEP 3 $\dfrac{\partial y}{\partial x}=-\dfrac{y}{x-2yz}=-\dfrac{1}{0-4}$; $\dfrac{\partial z}{\partial x}=-\dfrac{y}{3z^2-y^2}=-\dfrac{1}{12-1}$.

ANSWER $\boxed{\dfrac{\partial y}{\partial x}=\dfrac{1}{4},\quad \dfrac{\partial z}{\partial x}=-\dfrac{1}{11}}$

CHECK The lecture's point $(1,0,2)$ is not on the surface (see Exam trap).

S004·9 · T004a 00:18:39

W3.2 · $\partial z/\partial x$ and $\partial z/\partial y$ at the origin — ESA ★★

GIVEN $x^3+z^2+ye^{xz}+z\cos y=0$ at $(0,0,0)$.

STEP 1 $F_x=3x^2+zye^{xz}$, $F_y=e^{xz}-z\sin y$, $F_z=2z+xye^{xz}+\cos y$.

STEP 2 At the origin: $F_x=0$, $F_y=1$, $F_z=1$.

ANSWER $\boxed{\dfrac{\partial z}{\partial x}=0,\quad \dfrac{\partial z}{\partial y}=-1}$

CHECK The origin satisfies the equation.

S003·26 · T003b 00:09:15

W3.3 · Second derivative of an implicit function ESA ★★

GIVEN $x^x y^y z^z = c$. Show that at $x = y = z$, $\dfrac{\partial^2 z}{\partial x\,\partial y} = -[x\log(ex)]^{-1}$.

STEP 1 Take logs: $f = x\log x + y\log y + z\log z - \log c = 0$, so $f_x = 1 + \log x$, $f_y = 1 + \log y$, $f_z = 1 + \log z$.

STEP 2 $z_x = -\dfrac{1+\log x}{1+\log z}$, $z_y = -\dfrac{1+\log y}{1+\log z}$.

STEP 3 $z_{xy} = \dfrac{\partial}{\partial y} z_x = \dfrac{1+\log x}{(1+\log z)^2}\cdot\dfrac{1}{z}\,z_y$ (z depends on y).

STEP 4 At $x = y = z$: $z_y = -1$, so $z_{xy} = -\dfrac{1}{x(1+\log x)}$, and $1 + \log x = \log(ex)$.

ANSWER $\boxed{\dfrac{\partial^2 z}{\partial x\,\partial y} = -\dfrac{1}{x\log(ex)}}$

CHECK sympy residual 0.

ADDED FOR CLARITY ESA Jan-24 s3 1d · CW U1 P6

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.d · solved in W3.3

For the point on the surface $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -[x\log(ex)]^{-1}$.

SHORT-NOTE RECAP

For $f(x,y,z) = 0$, z is a function of x and y: $\dfrac{\partial z}{\partial x} = -\dfrac{f_x}{f_z}$ (hold y), $\dfrac{\partial z}{\partial y} = -\dfrac{f_y}{f_z}$, and $\dfrac{\partial y}{\partial x} = -\dfrac{f_x}{f_y}$ (hold z), provided the denominators are non-zero. Check that the point lies on the surface. For second derivatives, remember that z still depends on x, y.

4 Composite functions and the chain rule

If $u = f(x, y)$ and x, y depend on two new variables, u is a composite function of the new variables. Knowing this chain fixes the chain-rule formula: one term per intermediate variable. The exam questions in this topic prove identities that hold for any outer function f.

Key terms and definitions

Chain rule (several variables). If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables u, v, then $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$ and $\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. S003·13 · T003a 00:21:07

The chain of a problem. The list of dependencies, e.g. "u depends on x, y; x, y depend on r, t; so u is a composite function of r, t". Knowing it fixes which chain-rule formula to write. T004b 00:02:31

> *"We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v."* S003·13

Formulas

E4.1 · Chain rule, two intermediate variables ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

gives	Partial derivatives of a composite function.
applies	$z = f(x, y)$, $x = g(u, v)$, $y = h(u, v)$.
note	All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E4.2 · Chain rule, three intermediate variables ESA ★★★

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial v}{\partial t}\frac{\partial t}{\partial x}$$

gives	One term per intermediate variable.
applies	$v = f(r, s, t)$, each of r, s, t a function of x, y, z.
note	A term is 0 when that intermediate variable does not contain x.

S003·15 · T003a 00:23:10

E4.3 · Function of one intermediate variable ESA ★★

$$u = f(r),\ r = \sqrt{x^2+y^2+z^2} \implies \frac{\partial u}{\partial x} = f'(r)\,\frac{\partial r}{\partial x} = f'(r)\,\frac{x}{r}$$

GIVES $\partial u/\partial x$ in terms of $f'(r)$.

APPLIES u depends on x, y, z only through r.

S004·12 · T004b 00:05:07

Graphs and diagrams

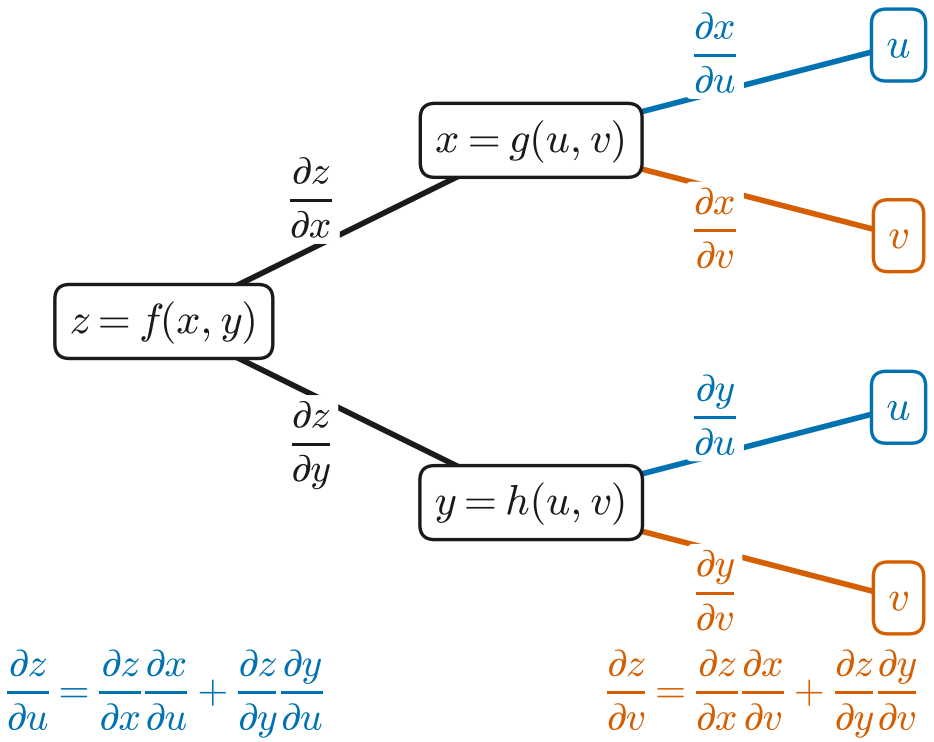
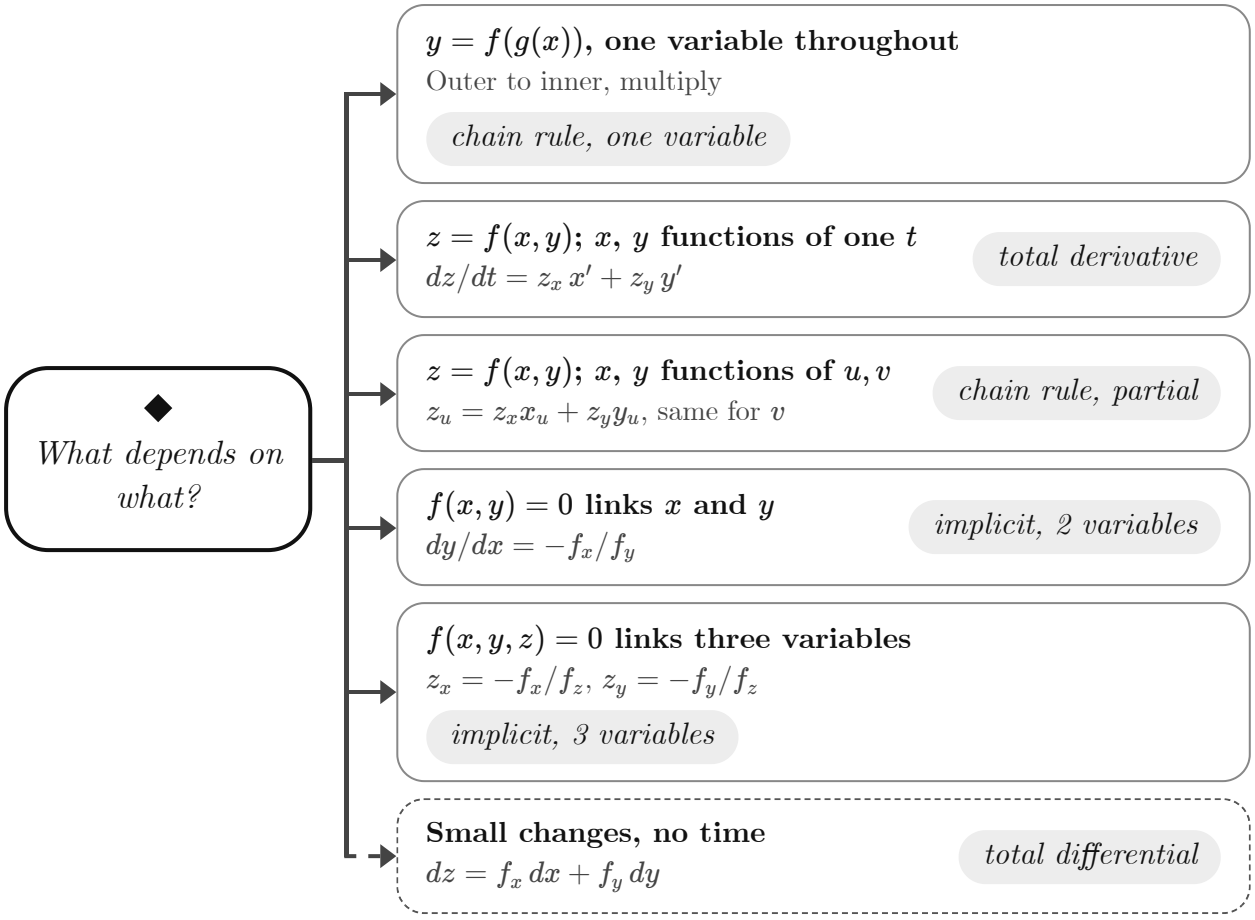


Figure G4.1. Tree diagram for the chain rule. ESA ★★★

S003·14 · T003a 00:19:42

How to read it. z branches to x and y, and each of those branches to both u and v, giving four leaves. For $\partial z/\partial u$ follow only the two paths ending at u (blue): $z_x x_u + z_y y_u$. For $\partial z/\partial v$ follow the two ending at v (vermilion). In an exam, draw the tree first and read the terms off it.

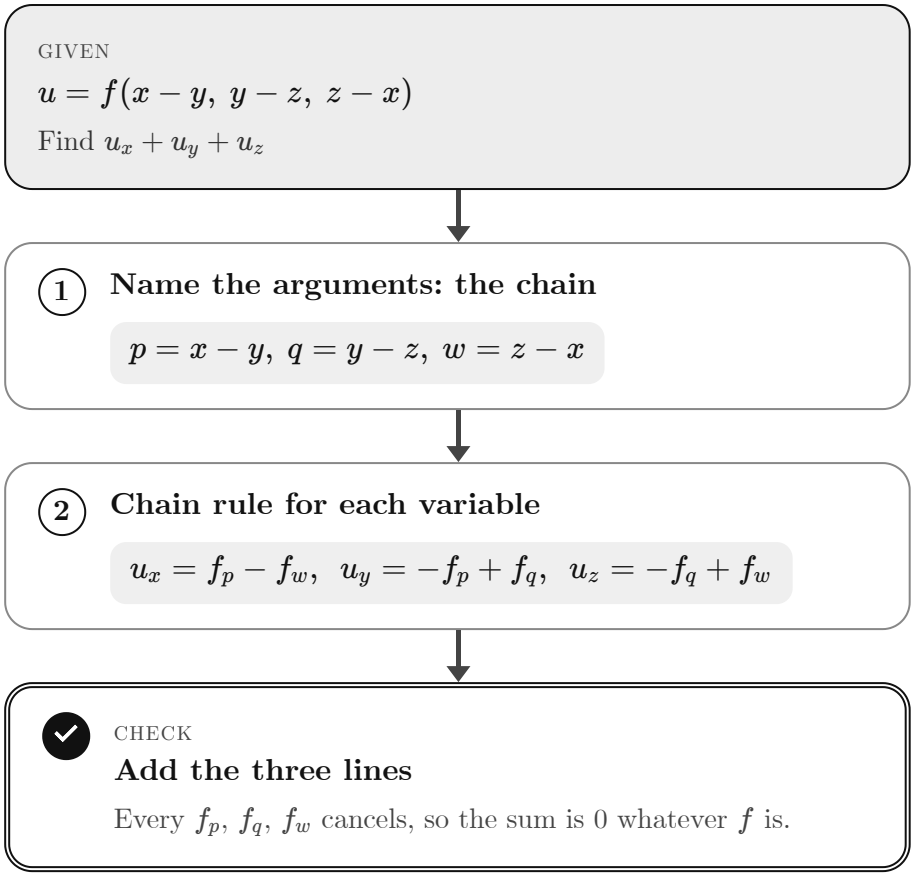
Flowcharts



Remark. The dashed route has no derivative with respect to a variable: it estimates a change. Everything else starts from the tree diagram.

Figure F4.1. Which rule? Read the chain first. ESA ★★★

S003·5 · S003·13 · S004·5 · S004·6 · T004b 00:02:31



Remark. Same method for $f(x/y, y/z, z/x)$ with weights x, y, z (W4.3): name the arguments, write each partial, then combine.

Figure F4.2. Proving $u_x + u_y + u_z = 0$ for $u = f(x-y,\ y-z,\ z-x)$. ESA ★★★

ESA Jan-24 s2 1d · T004b 00:02:31

Total derivative vs chain rule vs implicit differentiation

	Total derivative	Chain rule (partial)	Implicit function
Set-up	$z = f(x,y)$, $x(t)$, $y(t)$	$z = f(x,y)$, $x(u,v)$, $y(u,v)$	$f(x,y) = 0$ or $f(x,y,z) = 0$
Final variables	one: t	two: u, v	x (and y)
Formula	$z_x\, x' + z_y\, y'$	$z_x x_u + z_y y_u$	$-f_x/f_y$, $-f_x/f_z$
Result written	dz/dt	$\partial z/\partial u$, $\partial z/\partial v$	dy/dx, $\partial z/\partial x$
Typical question	rate of change at an instant	prove an identity	slope or tangent line

ADDED FOR CLARITY S003·5 · S003·13 · S004·4 · S004·6

Important points and why

1. It is essential to know the chain given in the problem: what depends on what. The formula follows from it. ESA ★★★ T004b 00:02:31
2. When u is a composite function of r, θ (or s, t), eliminate x and y from the final answer: you must know why you substitute. ESA ★★★ T004b 00:07:52 · T004b 00:10:28

Why does $u = f(x - y,\, y - z,\, z - x)$ give $u_x + u_y + u_z = 0$?
Each argument is a difference whose derivatives sum to zero: for $r = x - y$, $r_x + r_y + r_z = 1 - 1 + 0 = 0$. Adding the three chain-rule expressions gives $f_r(0) + f_s(0) + f_t(0)$. In words: moving x, y, z by the same amount changes none of the arguments.

ADDED FOR CLARITY ESA Jan-24 s2 1d

Why do the chain-rule answers keep f_r and f_θ?
The outer function f is unknown, so its partial derivatives cannot be computed; only the inner derivatives ($r_s = t$, $\theta_s = s/\sqrt{s^2 + t^2}$, ...) can.

S003·17 · T003a 00:26:28

Worked examples

W4.1 · $u = f(r)$ with r the distance — ESA ★★

GIVEN $u = f(r)$, $r = \sqrt{x^2 + y^2 + z^2}$.

STEP 1 $u_x = \dfrac{du}{dr}\, r_x = f'(r)\, r_x$.

STEP 2 $r_x = \dfrac{x}{\sqrt{x^2 + y^2 + z^2}} = \dfrac{x}{r}$.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} = f'(r)\, \dfrac{x}{r}}$

CHECK By symmetry $u_y = f'(r)\, y/r$, $u_z = f'(r)\, z/r$.

S004·12 · T004b 00:05:07

W4.2 · $u = \sin(xy) + y^3$, $x = e^s$, $y = s + t$ — ESA ★★

GIVEN $u = \sin(xy) + y^3$, $x = e^s$, $y = s + t$.

STEP 1 $u_x = y\cos(xy)$, $u_y = x\cos(xy) + 3y^2$; $x_s = e^s$, $y_s = 1$, $x_t = 0$, $y_t = 1$.

STEP 2 $u_s = y\cos(xy)e^s + (x\cos(xy) + 3y^2)$, $u_t = x\cos(xy) + 3y^2$, then substitute $x = e^s$, $y = s + t$.

ANSWER $\boxed{u_s = (s+t)e^s\cos\big(e^s(s+t)\big) + e^s\cos\big(e^s(s+t)\big) + 3(s+t)^2, \quad u_t = e^s\cos\big(e^s(s+t)\big) + 3(s+t)^2}$

CHECK sympy agrees.

S004·16 · T004b 00:09:32

W4.3 · $xu_x + yu_y + zu_z$ for $f(x/y,\ y/z,\ z/x)$ ESA ★★★

GIVEN $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$; find $xu_x + yu_y + zu_z$.

STEP 1 Let $p = \frac{x}{y}$, $q = \frac{y}{z}$, $w = \frac{z}{x}$.

STEP 2 $xu_x = x\left(\frac{1}{y} f_p - \frac{z}{x^2} f_w\right) = p\, f_p - w\, f_w$.

STEP 3 $yu_y = y\left(-\frac{x}{y^2} f_p + \frac{1}{z} f_q\right) = -p\, f_p + q\, f_q$.

STEP 4 $zu_z = z\left(-\frac{y}{z^2} f_q + \frac{1}{x} f_w\right) = -q\, f_q + w\, f_w$.

STEP 5 Add the three lines.

ANSWER $$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0}$$

CHECK u is homogeneous of degree 0 (scaling x, y, z leaves every ratio unchanged), so Euler's theorem gives 0 too.

ADDED FOR CLARITY ESA Jan-24 s1 1d

W4.4 · $u_x + u_y + u_z$ for $f(x - y,\ y - z,\ z - x)$ ESA ★★★

GIVEN $u = f(x - y,\ y - z,\ z - x)$; find $u_x + u_y + u_z$.

STEP 1 Let $p = x - y$, $q = y - z$, $w = z - x$.

STEP 2 $u_x = f_p - f_w$, $u_y = -f_p + f_q$, $u_z = -f_q + f_w$.

STEP 3 Add.

ANSWER $$\boxed{\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0}$$

CHECK The three derivatives of each argument add to zero, so every f-term cancels.

ADDED FOR CLARITY ESA Jan-24 s2 1d

W4.5 · $z = f(u, v)$, $u = x^2 - y^2$, $v = 2xy$ ESA ★★★

GIVEN Show $z_x^2 + z_y^2 = 4(u^2 + v^2)^{1/2}\,(z_u^2 + z_v^2)$.

STEP 1 $z_x = z_u(2x) + z_v(2y)$, $z_y = z_u(-2y) + z_v(2x)$.

STEP 2 $z_x^2 + z_y^2 = 4(x^2 + y^2)(z_u^2 + z_v^2)$: the cross terms $8xy\, z_u z_v$ cancel.

STEP 3 $u^2 + v^2 = (x^2 - y^2)^2 + 4x^2y^2 = (x^2 + y^2)^2$, so $x^2 + y^2 = (u^2 + v^2)^{1/2}$.

ANSWER $$\boxed{z_x^2 + z_y^2 = 4\sqrt{u^2 + v^2}\,(z_u^2 + z_v^2)}$$

CHECK Numerical check with a sample f in sympy: residual $\approx 10^{-14}$.

ADDED FOR CLARITY ESA Jan-24 s3 1b

W4.6 · $z = f(x,y),\ x = e^u \sin v,\ y = e^u \cos v$ ESA ★★★

GIVEN Show $z_x^2 + z_y^2 = e^{-2u}(z_u^2 + z_v^2)$.

STEP 1 $z_u = z_x\, e^u \sin v + z_y\, e^u \cos v = x z_x + y z_y$.

STEP 2 $z_v = z_x\, e^u \cos v - z_y\, e^u \sin v = y z_x - x z_y$.

STEP 3 $z_u^2 + z_v^2 = (x^2 + y^2)(z_x^2 + z_y^2)$: the cross terms cancel.

STEP 4 $x^2 + y^2 = e^{2u}$.

ANSWER $\boxed{z_x^2 + z_y^2 = e^{-2u}\,(z_u^2 + z_v^2)}$

CHECK sympy with a sample f: exact 0.

ADDED FOR CLARITY ESA Jan-24 s4 1b

W4.7 · $z = \log(u^2 + v)$ ESA ★★★

GIVEN $z = \log(u^2+v)$, $u = e^{x+y^2}$, $v = x + y^2$; find $2y\, z_x - z_y$.

STEP 1 $z_u = \dfrac{2u}{u^2+v}$, $z_v = \dfrac{1}{u^2+v}$; $u_x = u$, $u_y = 2yu$, $v_x = 1$, $v_y = 2y$.

STEP 2 $z_x = \dfrac{2u^2+1}{u^2+v}$, $z_y = \dfrac{2y(2u^2+1)}{u^2+v}$.

ANSWER $\boxed{2y\,\dfrac{\partial z}{\partial x} - \dfrac{\partial z}{\partial y} = 0}$

CHECK CW answer: 0. Both depend on x, y only through $x + y^2$.

ESA Jun-24 1a · CW U1 P14

Past-paper questions

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W4.3

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q1.d · solved in W4.4

If $u = f(x-y,\, y-z,\, z-x)$, find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.b · solved in W4.5

If $z = f(u,v)$ and $u = x^2 - y^2$, $v = 2xy$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2+v^2)^{1/2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.b · solved in W4.6

If $z = f(x,y)$ and $x = e^u \sin v$, $y = e^u \cos v$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA June 2024 | 6 marks | UE23MA141A Q1.a · solved in W4.7

If $z = \log(u^2 + v)$, where $u = e^{x+y^2}$, $v = x + y^2$, then find the value of $2y\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$.

SHORT-NOTE RECAP

If $u = f(x, y)$ and x, y depend on s, t (or r, θ), u is a composite function of s, t, and $u_s = f_x\, x_s + f_y\, y_s$, $u_t = f_x\, x_t + f_y\, y_t$: one term per intermediate variable. Write the chain first. If f is unknown, keep f_x, f_y as symbols. Eliminate x, y from the final answer. Standard results: $u_x + u_y + u_z = 0$ for $f(x - y, y - z, z - x)$; $xu_x + yu_y + zu_z = 0$ for $f(x/y, y/z, z/x)$.

5 Composite functions in polar coordinates

Most of the class 004 problems use $x = r\cos\theta$, $y = r\sin\theta$ (or the reverse, $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$). Compute the inner derivatives once, apply the chain rule, then eliminate x and y. Functions of x^2+y^2 alone give $u_\theta = 0$; $\tan^{-1}(y/x)$ gives $u_\theta = 1$.

Formulas

E5.1 · Polar coordinates: derivatives you reuse

ESA ★★

$$\begin{aligned} &x = r\cos\theta,\ y = r\sin\theta: && x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta \\ &r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: && r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2} \end{aligned}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r,\theta)$.

S004·13 · S004·15 · S004·17

E5.2 · Polar identity

ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

NOTE Proved in W5.6.

CW U1 P15

Graphs and diagrams

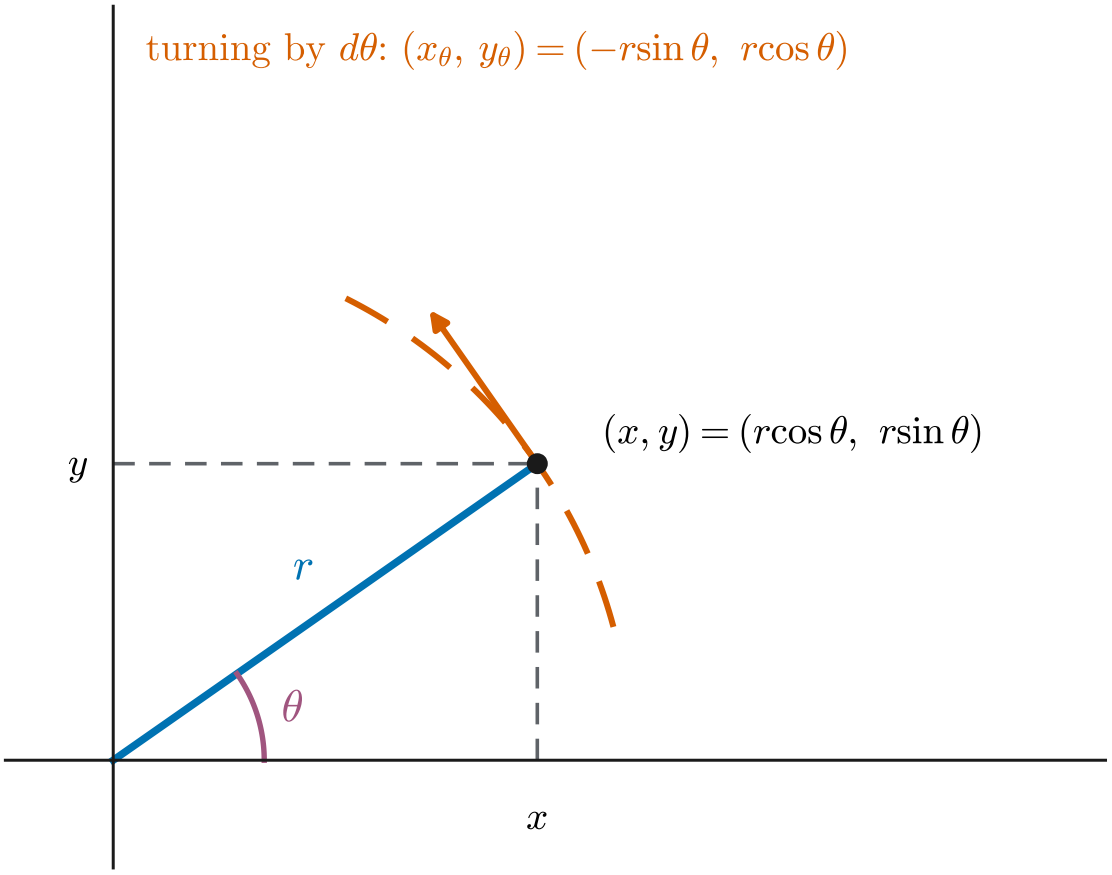
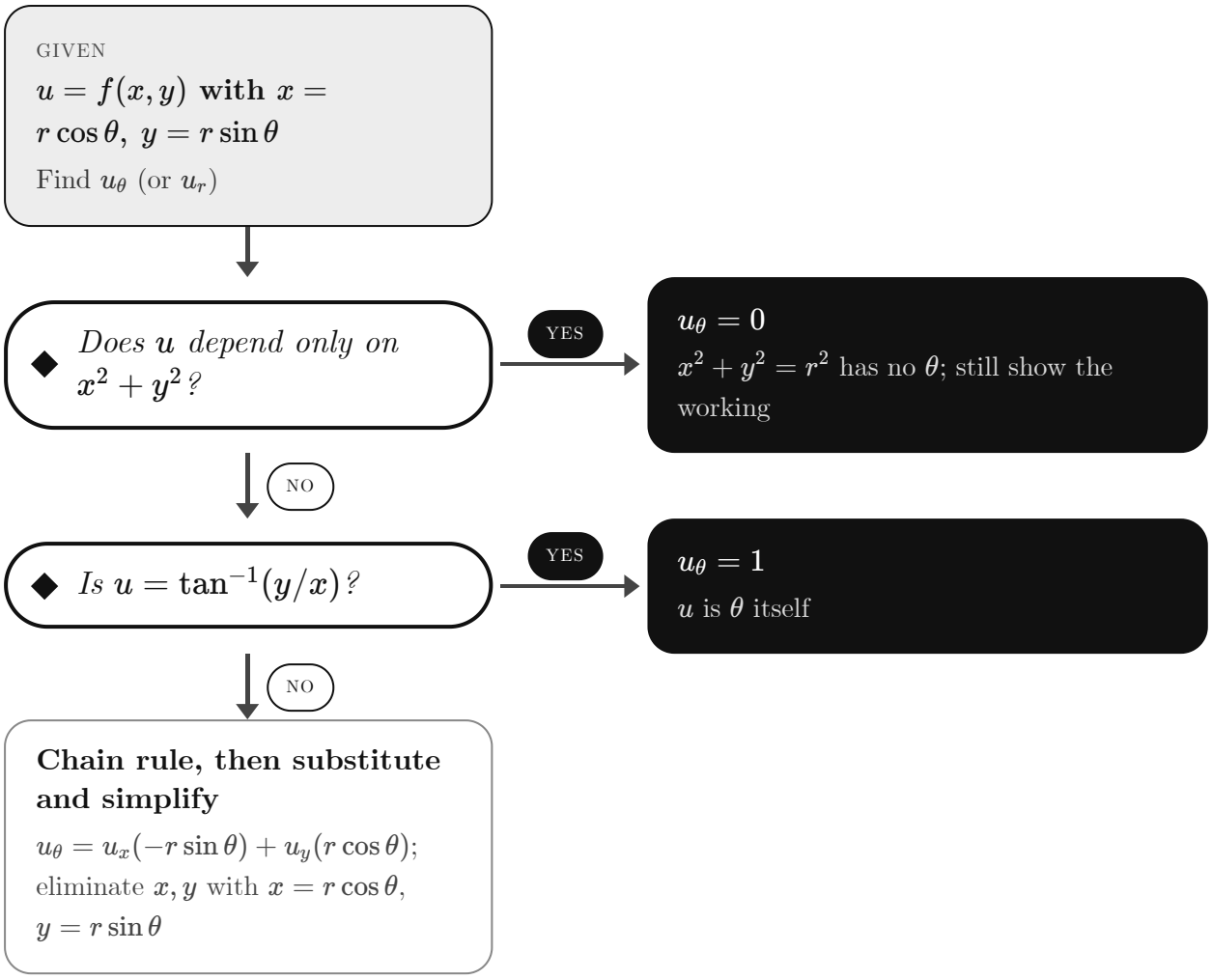


Figure G5.1. Polar coordinates and the inner derivatives. ESA ★★

ADDED FOR CLARITY S004·10 · S004·13 · S004·17

How to read it. A point at distance r and angle θ has $x = r\cos\theta$, $y = r\sin\theta$. Turning by $d\theta$ moves it along the dashed circle, so $x_\theta = -r\sin\theta$ and $y_\theta = r\cos\theta$, while r stays fixed. That is why any function of $x^2 + y^2 = r^2$ alone has $u_\theta = 0$.

Flowcharts



Remark. The first two rungs let you predict the answer; the exam still expects the full chain-rule working (W5.1, W5.3, W5.4).

Figure F5.1. Solving a polar-coordinate composite problem. ESA ★★

S004·10 · S004·11 · S004·13 · S004·15 · T004b 00:07:52

Important points and why

1. Functions of $x^2 + y^2$ alone (e.g. $\sin(x^2 + y^2)$, $e^{x^2+y^2}$) have $\partial u/\partial\theta = 0$ in polar coordinates, since $x^2 + y^2 = r^2$ does not involve θ. ESA ★★ ADDED FOR CLARITY S004·10 · S004·13 · S004·14
2. $\tan^{-1}(y/x)$ is exactly θ in polar coordinates, so $\partial u/\partial\theta = 1$. ESA ★★ ADDED FOR CLARITY S004·15

Exam trap. Do not stop with x and y still in an answer that should be in r, θ or s, t. T004b 00:07:52

Worked examples

W5.1 · $\sin(x^2+y^2)$ in polar form — ESA ★★

given $u=\sin(x^2+y^2)$, $x=r\cos t$, $y=r\sin t$; find u_t.

step 1 $u_x=2x\cos(x^2+y^2)$, $u_y=2y\cos(x^2+y^2)$; $x_t=-r\sin t$, $y_t=r\cos t$.

step 2 $u_t=2\cos(r^2)\,[\,r\cos t(-r\sin t)+r\sin t(r\cos t)\,]$.

answer $\boxed{\dfrac{\partial u}{\partial t}=0}$

check $u=\sin r^2$ does not involve t at all.

S004·10 · T004b 00:00:10

W5.2 · Three intermediate variables — ESA ★★

given $u=x^2y+yz^3$, $x=r\cos\theta$, $y=r\sin\theta$, $z=\theta$; find u_θ.

step 1 $u_x=2xy$, $u_y=x^2+z^3$, $u_z=3yz^2$; $x_\theta=-r\sin\theta$, $y_\theta=r\cos\theta$, $z_\theta=1$.

step 2 $u_\theta=2xy(-r\sin\theta)+(x^2+z^3)(r\cos\theta)+3yz^2$.

step 3 Substitute: $-2r^3\cos\theta\sin^2\theta+r^3\cos^3\theta=r^3\cos\theta(3\cos^2\theta-2)$.

answer $\boxed{\dfrac{\partial u}{\partial\theta}=r^3\cos\theta\,(3\cos^2\theta-2)+r\,(3\theta^2\sin\theta+\theta^3\cos\theta)}$

check sympy agrees.

S004·11 · T004b 00:03:09

W5.3 · $e^{x^2+y^2}$ in polar form — ESA ★★

given $u=e^{x^2+y^2}$, $x=r\cos\theta$, $y=r\sin\theta$.

step 1 $u_\theta=2xe^{x^2+y^2}(-r\sin\theta)+2ye^{x^2+y^2}(r\cos\theta)$.

step 2 $=2re^{r^2}(-r\cos\theta\sin\theta+r\sin\theta\cos\theta)$.

answer $\boxed{\dfrac{\partial u}{\partial\theta}=0}$

check $u=e^{r^2}$ has no θ.

S004·13 · S004·14 · T004b 00:06:38

W5.4 · Show $\partial u/\partial\theta = 1$ ESA ★★

GIVEN $u = \tan^{-1}(y/x)$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_x = \dfrac{-y}{x^2+y^2}$, $u_y = \dfrac{x}{x^2+y^2}$; $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$.

STEP 2 $u_\theta = \dfrac{r^2\sin^2\theta + r^2\cos^2\theta}{r^2}$.

ANSWER $\boxed{\dfrac{\partial u}{\partial\theta} = 1}$

CHECK $u = \tan^{-1}(\tan\theta) = \theta$.

S004·15 · T004b 00:08:40

W5.5 · $u = f(r,\theta)$ from Cartesian coordinates ESA ★★

GIVEN $u = f(r,\theta)$, $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

STEP 1 $r_x = \dfrac{x}{r}$, $r_y = \dfrac{y}{r}$, $\theta_x = -\dfrac{y}{r^2}$, $\theta_y = \dfrac{x}{r^2}$.

STEP 2 $u_x = f_r r_x + f_\theta\theta_x$, $u_y = f_r r_y + f_\theta\theta_y$.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} = f_r\dfrac{x}{r} - f_\theta\dfrac{y}{r^2}, \quad \dfrac{\partial u}{\partial y} = f_r\dfrac{y}{r} + f_\theta\dfrac{x}{r^2}}$

S004·17 · T004b 00:10:47

W5.6 · Polar identity ESA ★★

GIVEN $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_r = u_x\cos\theta + u_y\sin\theta$.

STEP 2 $u_\theta = -u_x\, r\sin\theta + u_y\, r\cos\theta$, so $\frac{1}{r}u_\theta = -u_x\sin\theta + u_y\cos\theta$.

STEP 3 Square and add: the cross terms cancel and $\sin^2\theta + \cos^2\theta = 1$.

ANSWER $\boxed{u_x^2 + u_y^2 = u_r^2 + \dfrac{1}{r^2}\,u_\theta^2}$

CHECK sympy with a sample u: residual $\approx 10^{-16}$.

CW U1 P15

SHORT-NOTE RECAP

With $x = r\cos\theta$, $y = r\sin\theta$: $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$; with $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$: $r_x = x/r$, $r_y = y/r$, $\theta_x = -y/r^2$, $\theta_y = x/r^2$. Functions of x^2+y^2 alone have $u_\theta = 0$, $\tan^{-1}(y/x)$ has $u_\theta = 1$, and $u_x^2 + u_y^2 = u_r^2 + u_\theta^2/r^2$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
$f(x, y) = c$	Level curve / implicit relation	Defines y as a function of x	E2.1
$f(x, y, z) = 0$	Level surface	Defines z as a function of x, y	E3.1
c, c_1	Level	The constant value of f on one level surface	E2.1
f_x, f_y, f_z	Partial derivatives of the relation	Numerator and denominator of the implicit formulas	E2.1, E3.1
df, dx, dy	Differentials	$df = f_x\,dx + f_y\,dy$; zero on a level curve	E1.1
$\dfrac{dy}{dx}$	Slope of a level curve	$-f_x / f_y$	E2.1, E2.2
m, (x_0, y_0)	Slope, point of contact	Tangent $y - y_0 = m(x - x_0)$	E2.2
$u = f(x, y)$	Composite function	Outer function of the chain (unknown f kept as f_x, f_y)	E4.1
s, t / u, v	New variables	Final variables of the composite function	E4.1
p, q, w	Named arguments	e.g. $p = x - y$, $q = y - z$, $w = z - x$	E4.2
r, θ	Polar coordinates	$x = r\cos\theta$, $y = r\sin\theta$	E5.1, E5.2
r (slide 004·12)	Distance from the origin	$\sqrt{x^2 + y^2 + z^2}$	E4.3
$f'(r)$	Ordinary derivative	u depends on x, y, z only through r	E4.3
$\ln$, $\log$	Natural logarithm	Appears from a^x and from x^x	E2.3

B Sources, coverage and gaps

Source	Path	Use
Deck 004 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.*	18 slides; content on 3–17
Transcript 004 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.*	22.4 min lecture
Transcript 004 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.*	11.7 min problems
Deck 003, slides 26–27, and transcript 003 part II (end)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.* · AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.*	Slides 26–27: two implicit-function problems; slides 5–17: the definitions class 004 builds on
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 6, 14–18 (with answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Videos 004 (2 .mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_*.mp4	Not opened: every gap was settled from the slides, transcripts and algebra

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. In scope: one implicit-function question (Jan 2024 set 3 Q1.d, 6 marks) and five composite-function (chain rule) questions (Jan 2024 set 1 Q1.d and set 2 Q1.d, 5 marks; set 3 Q1.b, set 4 Q1.b and June 2024 Q1.a, 6 marks). Polar-coordinate problems and tangent lines to implicit curves were not asked in these papers; they are computed in class and in CW problems 15–18. Total-derivative and rate problems (class 003) are left out.

Known gaps, conflicts and corrections

1. Slide 004·4 prints $dy/dx = -f_y/f_x$ for a level curve. It should be $-f_x/f_y$, as slide 004·5's branch diagram and slide 004·6 show.
2. Slide 004·3's numerator is hard to read in the rendering. The correct dy/dx for $x^3y + 3xy^2 = 3$ is $-3y(x^2 + y)/(x(x^2 + 6y))$.
3. In the lecture (T004a 00:18:39) problem 3 of deck 004 is worked at the point (1, 0, 2), giving 0 for both derivatives. That point is not on $z^3 + xy - y^2z = 6$ (it gives 8). The slide's point (0, 1, 2) is on the surface and gives $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$.

4. The lecture says the differential is "minus z_x by z_y" (T004a 00:07:05); that expression is the slope dy/dx, not the differential. The differential is $df = f_x\,dx + f_y\,dy$.
5. Slides 004·4, 004·6 and 004·7 show screenshots from online 3-D graphers (calculus7.com, intmath.com). They are redrawn as Figures G1.1 and G2.2 rather than copied.
6. The same class 004 content is also in the classes 003–004 pack; problems and answers are identical in both.
7. Transcript recognition errors were corrected silently: "Gosteep" → cos ϑ, "composive" → composite, "dou dita" → $\partial\vartheta$, "sec y" → z_y, "psi x" → sin x.

Coverage

Source	Citations	Slides cited (slide × citations)
S003	19	5×2, 7×1, 8×1, 13×5, 14×2, 15×1, 17×1, 26×3, 27×3
S004	68	3×6, 4×12, 5×7, 6×8, 7×4, 8×3, 9×5, 10×4, 11×2, 12×2, 13×5, 14×2, 15×4, 16×1, 17×3
CW	9	
ESA	9	
T003a	6	
T003b	6	
T004a	41	
T004b	16	

Slides 004·1, 004·2 and 004·18 are title, session and closing slides with no content. From class 003, slides 26–27 (implicit-function problems) are worked in full; its slides 5–17 are cited only where class 004 reuses class 003's definitions (total differential, chain rule, tree diagram).